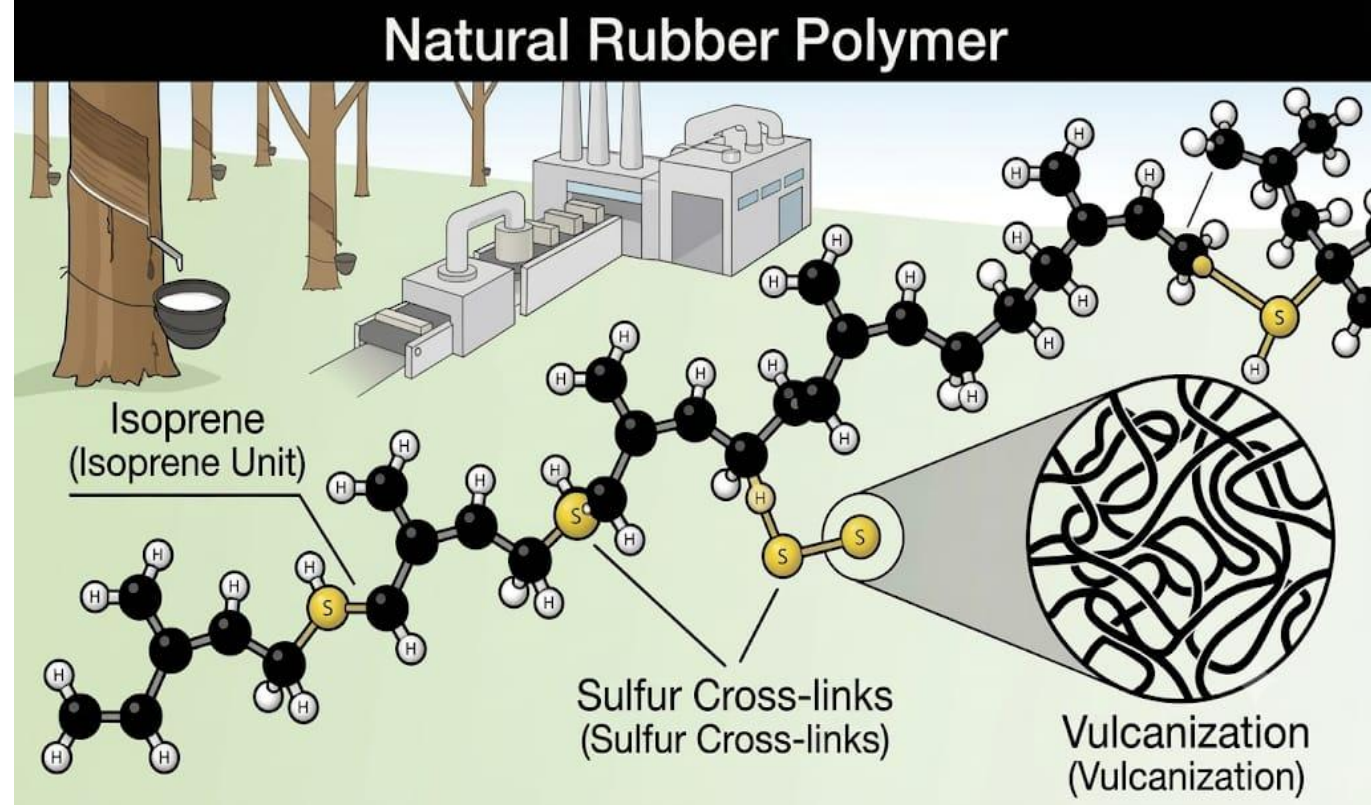


RUBBER ***PRODUCTION***



INTRODUCTION:

- ☐ Natural production
- ☐ Rubber tree
- ☐ Polymer
- ☐ Manual collection processes
- ☐ Biodegradable materials
- ☐ Elasticity(not strong)
- ☐ Add carbon and sulfur to strong

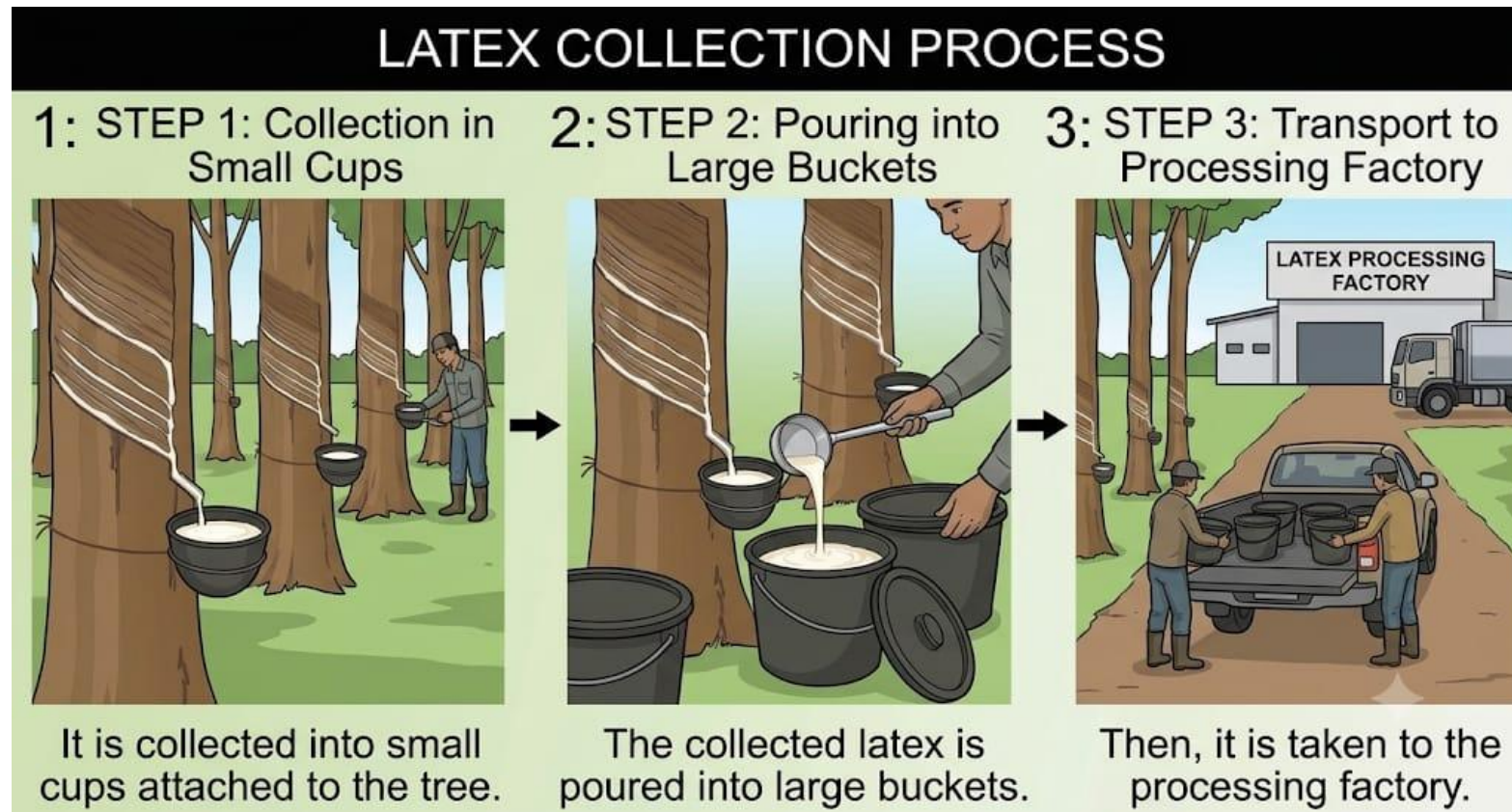


First step : Tapping

- ❖ Fresh latex is collected from rubber trees.
- ❖ Workers make small cuts on the tree bark.
- ❖ The liquid latex flows out slowly.

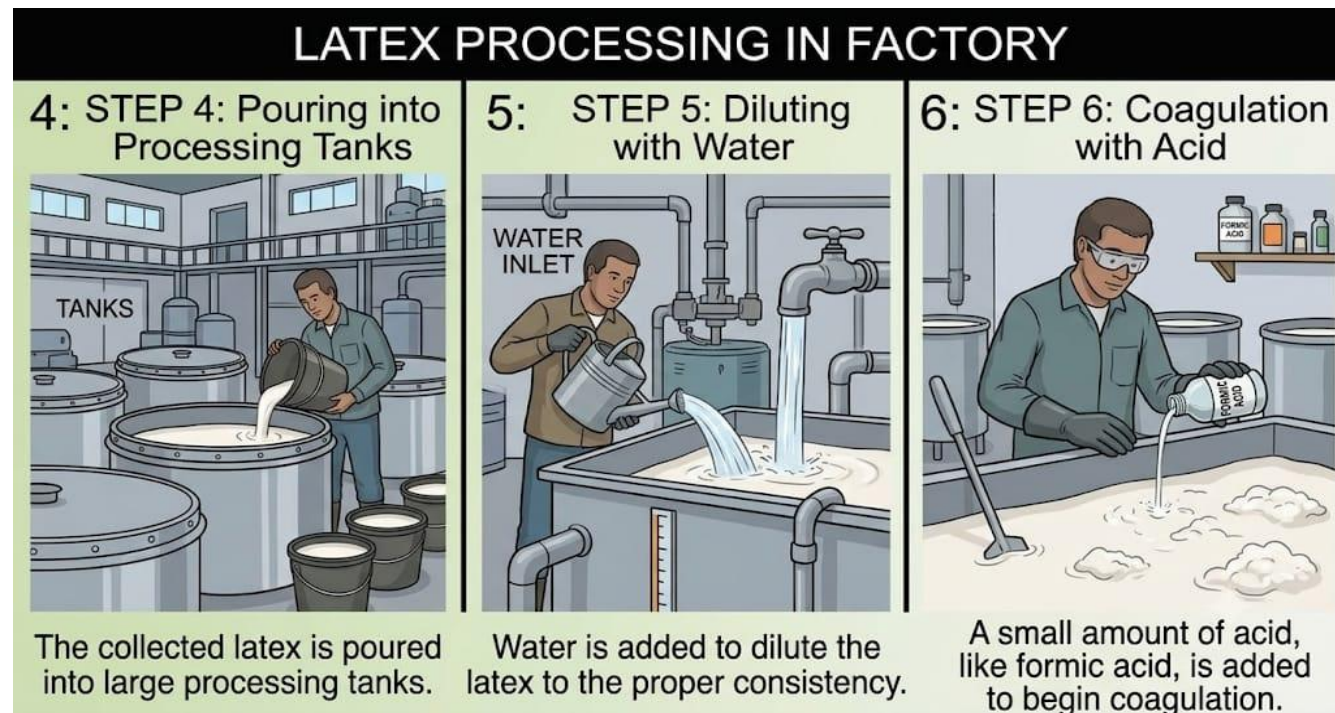


- ❖ It is collected into small cups attached to the tree.
- ❖ The collected latex is poured into large buckets.
- ❖ Then, it is taken to the processing factory.

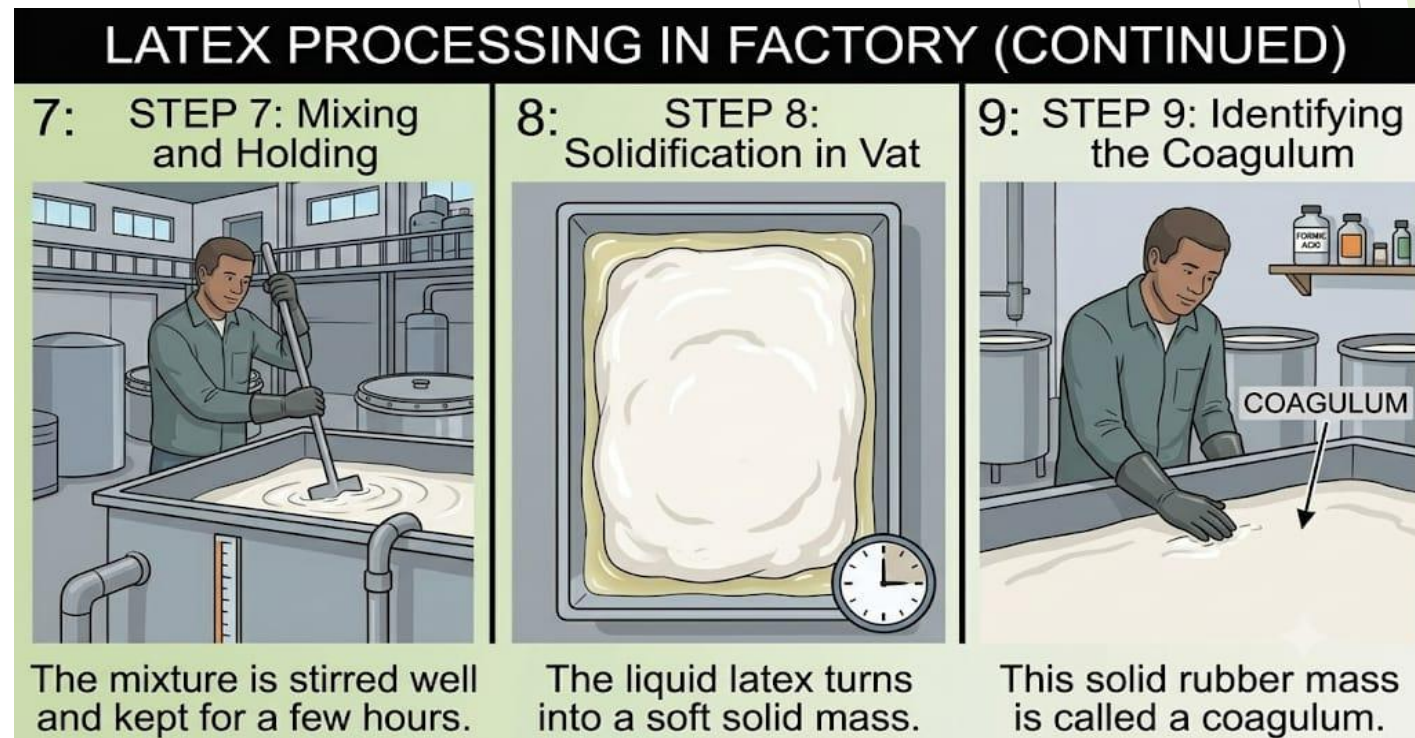


Second step : Coagulation

- ❖ The collected latex is poured into processing tanks.
- ❖ Water is added to dilute the latex.
- ❖ A small amount of acid, like formic acid, is added.

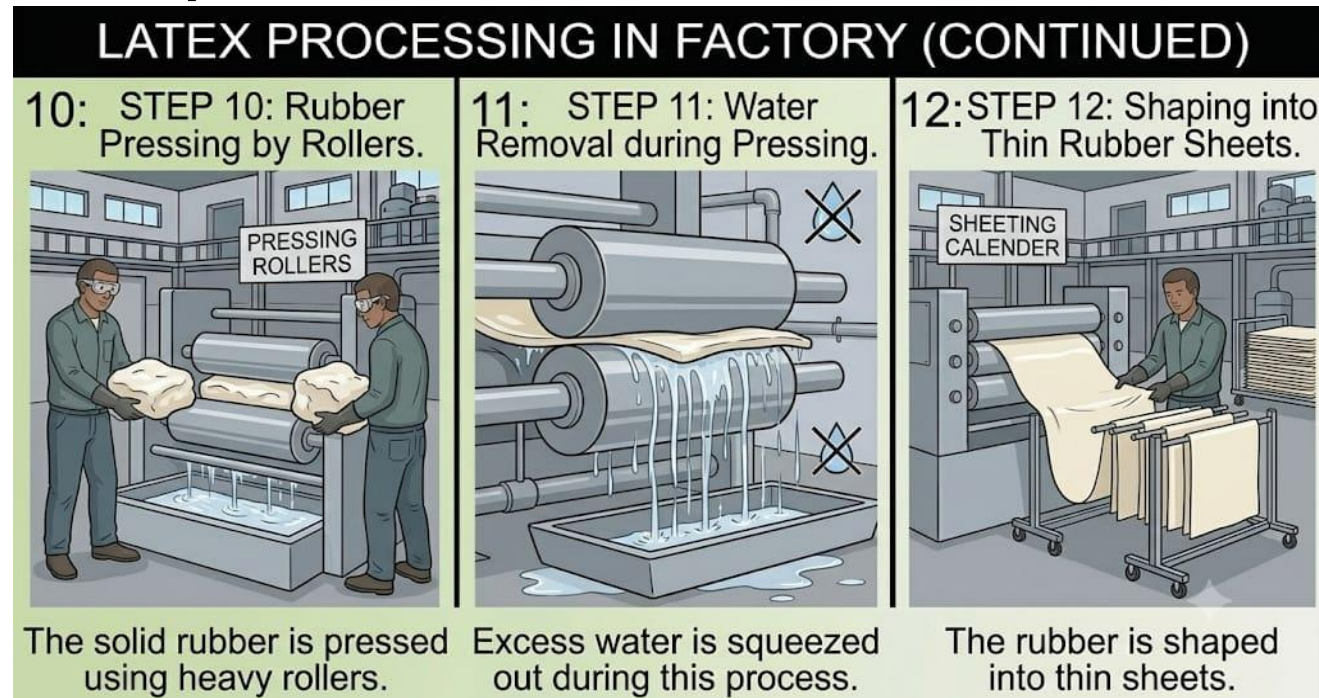


- ❖ The mixture is stirred well and kept for a few hours.
- ❖ The liquid latex turns into a soft solid mass.
- ❖ This solid rubber mass is called a coagulum.

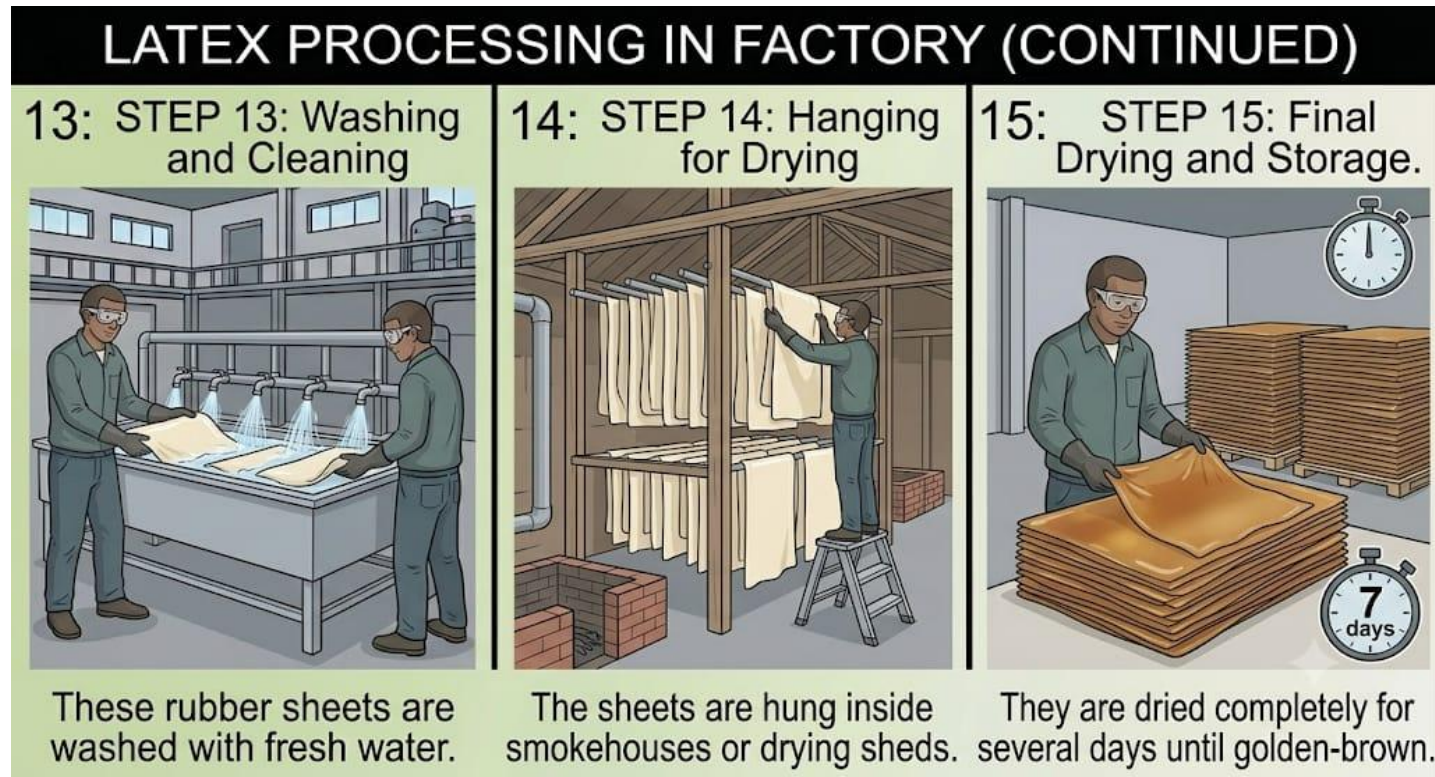


Third step : Milling and Drying

- ❖ The solid rubber is pressed using heavy rollers.
- ❖ Excess water is squeezed out during this process.
- ❖ The rubber is shaped into thin sheets.

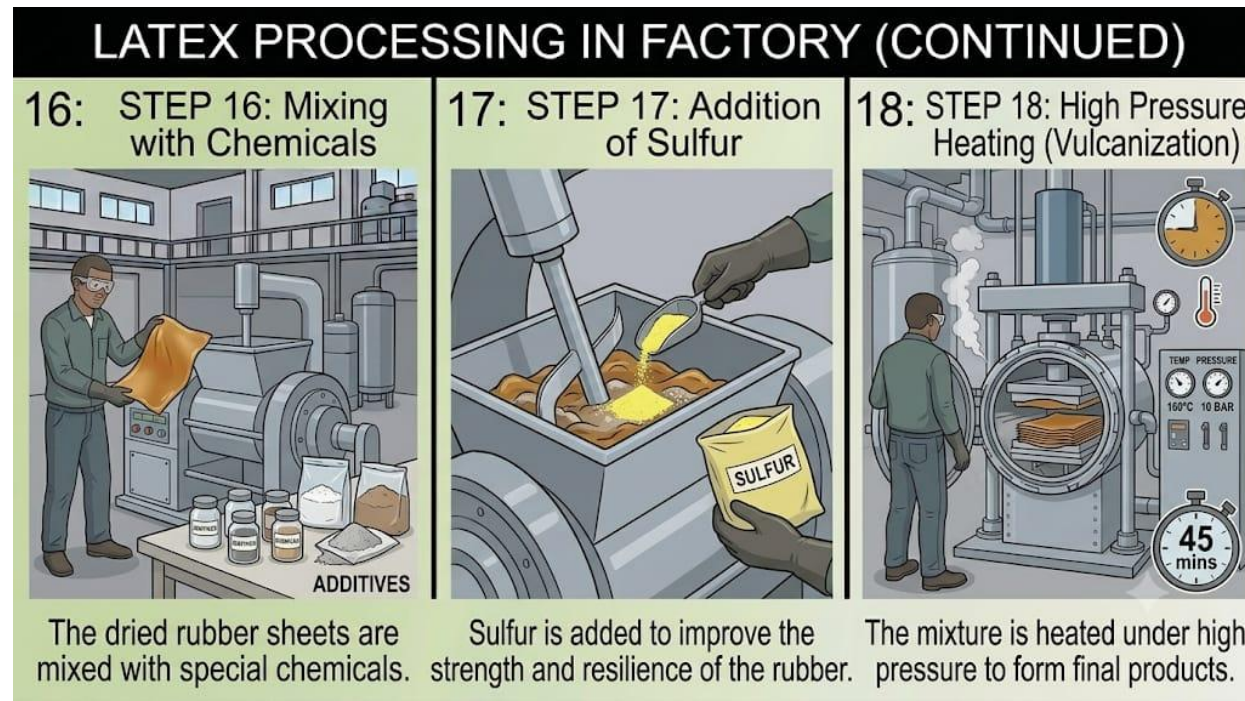


- ❖ These rubber sheets are washed with fresh water.
- ❖ The sheets are hung inside smokehouses or drying sheds.
- ❖ They are dried completely for several days.



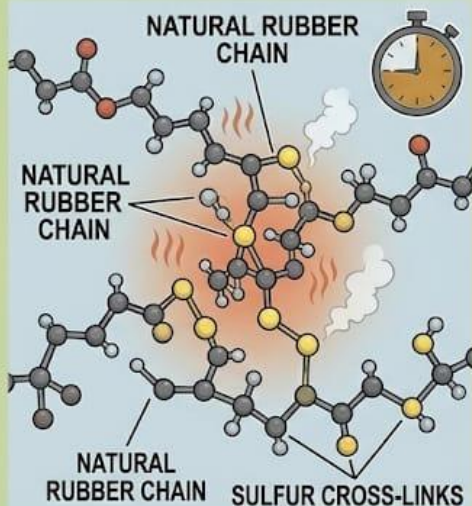
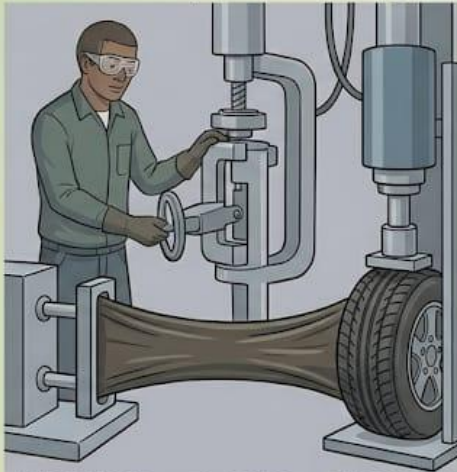
Fourth step : Compounding and Vulcanization

- ❖ The dried rubber sheets are mixed with special chemicals.
- ❖ Sulfur is added to improve the strength of the rubber.
- ❖ The mixture is heated under high pressure.



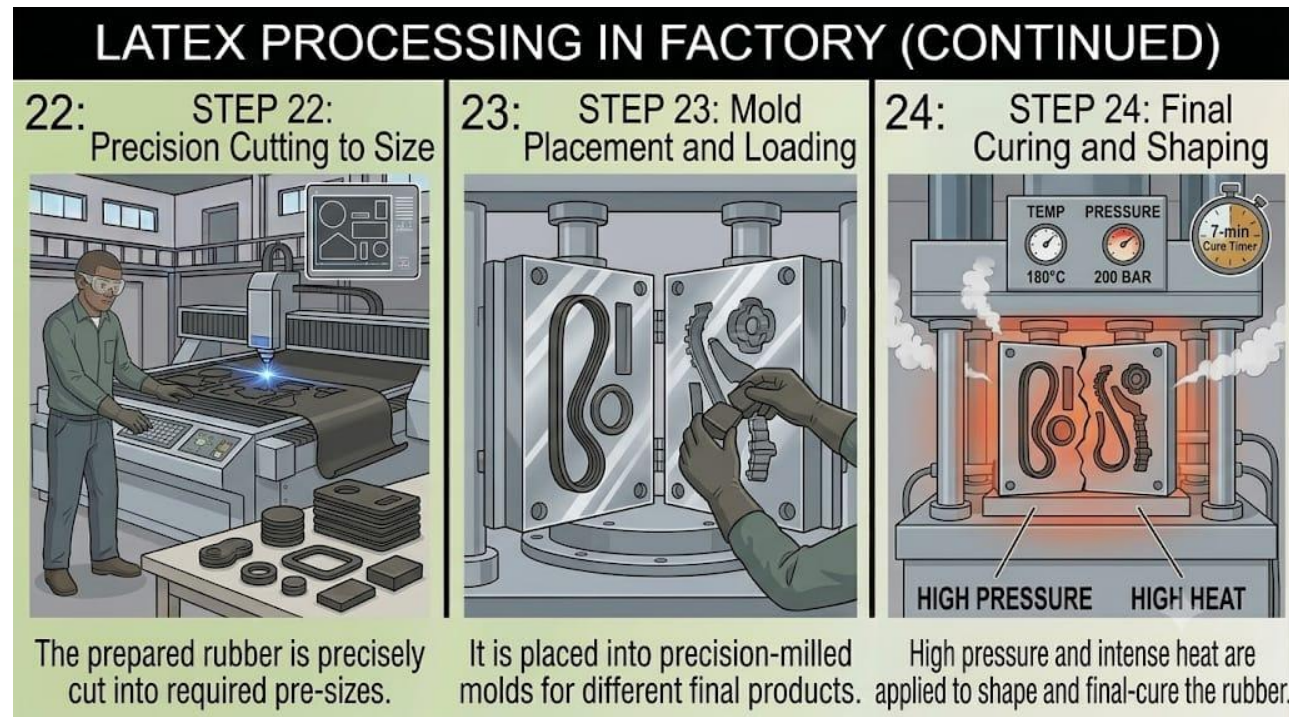
- ❖ This heating process is called vulcanization.
- ❖ It makes the rubber stronger, elastic, and durable.
- ❖ The rubber becomes heat-resistant after vulcanization.

LATEX PROCESSING IN FACTORY (CONTINUED)

19: STEP 19: The Vulcanization Phenomenon  The diagram illustrates the molecular process of vulcanization. It shows several 'NATURAL RUBBER CHAIN' molecules, which are long, coiled chains of carbon (black) and hydrogen (white) atoms. These chains are being heated, as indicated by wavy lines and a stopwatch icon. Sulfur atoms (yellow) are introduced, and 'SULFUR CROSS-LINKS' are formed between the chains, creating a more rigid, interconnected network.	20: STEP 20: Mechanical Property Testing  A worker wearing safety glasses and gloves is operating a machine that stretches a large rubber tire. The machine is labeled with the words 'STRONGER', 'ELASTIC', and 'DURABLE'.	21: STEP 21: Heat Resistance Application  A rubber hose is being tested with a blowtorch. A flame is applied to the hose, and a sign with a flame icon and the words 'HEAT RESISTANT' is shown next to it.
This molecular heating process is called vulcanization.	It makes the rubber stronger, elastic, and durable.	The rubber becomes heat-resistant after vulcanization.

Fifth step : **Molding and Manufacturing**

- ❖ The prepared rubber is cut into required sizes.
- ❖ It is placed into molds of different products.
- ❖ High pressure and heat are applied to shape the rubber.



- ❖ Items like tires, gloves, and shoes are manufactured.
- ❖ The finished products are checked for quality control.
- ❖ Finally, the products are packaged and sent to the market.



**THANK
YOU...!**